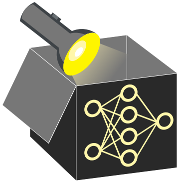
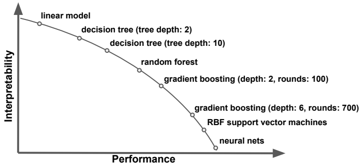
Interpretable Machine Learning



Intro to IML

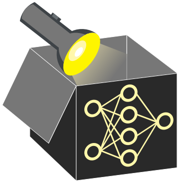
Introduction, Motivation, and History



Learning goals

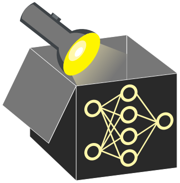
- Why interpretability?
- Developments until now?
- Use cases for interpretability

WHY INTERPRETABILITY?



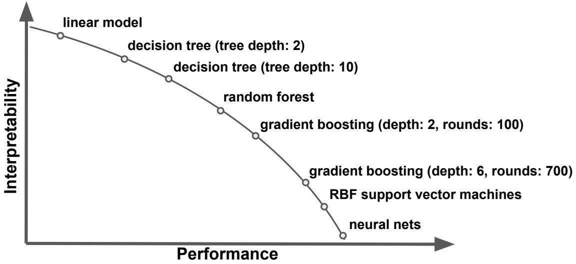
- ML aids decision-making in applications through strong predictive performance
- ML models are black boxes, e.g., XGBoost, RBF SVM, or DNNs
  $\rightsquigarrow$ too complex to be understood by humans
- Some applications are “learn to understand”, not “learn to predict”

WHY INTERPRETABILITY?

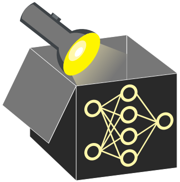


- ML aids decision-making in applications through strong predictive performance
- ML models are black boxes, e.g., XGBoost, RBF SVM, or DNNs
  $\rightsquigarrow$ too complex to be understood by humans
- Some applications are “learn to understand”, not “learn to predict”
- When deploying ML models, lack of explanations
  1. reduces trust
  2. creates barriers to adoption

$\rightsquigarrow$ Many disciplines with required trust rely on traditional models, e.g., linear models, despite lower predictive performance

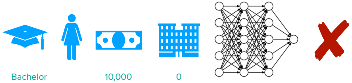


INTERPRETABILITY IN HIGH-STAKES DECISIONS



Examples of critical areas where ML-based decisions can affect human life

- Credit scoring and insurance applications ▸ "Society of Actuaries" 2021
  - Reasons for not granting a loan
  - Fraud detection in insurance claims



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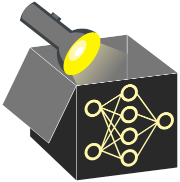
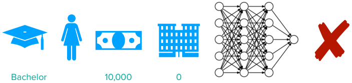
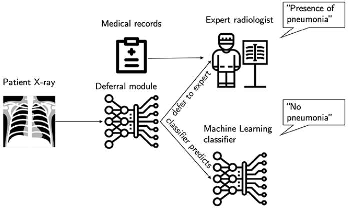


Examples of critical areas where ML-based decisions can affect human life

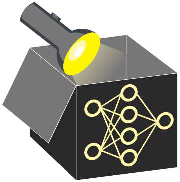
- Credit scoring and insurance applications ▸ "Society of Actuaries" 2021
  - Reasons for not granting a loan
  - Fraud detection in insurance claims
- Medical applications
  - Identification of diseases
  - Recommendations of treatments
- . . .





▸ Click for source

NEED FOR INTERPRETABILITY

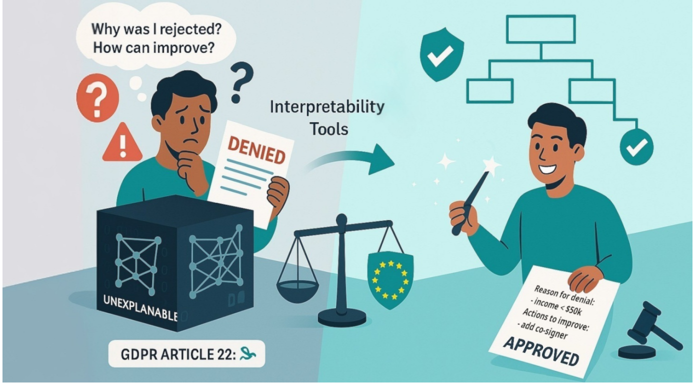


Need for interpretability becoming increasingly important from a legal perspective

- General Data Protection Regulation (GDPR) requires for some applications that models have to be explainable ▸ "Goodman and Flaxman" 2017

⇝ *EU Regulations on Algorithmic Decision-Making and a "Right to Explanation"*

- *Ethics guidelines for trustworthy AI* ▸ "European Commission" 2019



BRIEF HISTORY OF INTERPRETABILITY

- 18th/19th century:
  Linear models (Gauss, Legendre, Quetelet)
- Mid-20th century:
  Emergence of sensitivity analysis
- Mid- to late 20th century:
  Rule-based ML, incl. decision rules and trees
- Early 2000s:
  Interpretation tools for ensembles:
  - variable importance for random forests
  - partial dependence for gradient boosting
- Early 2010s:
  Explainability methods for deep learning
- Late 2010s:
  IML / XAI as a distinct research area

▶ "Carl Friedrich Gauss" 1828

▶ "Wikipedia" 1777